

MATH3026/4022: Time Series Analysis/Time Series and Forecasting
Exercises Sheet 3 – SOLUTIONS

1. Consider the process

$$X_t = Z_t + \frac{1}{4}Z_{t-1} - \frac{1}{8}Z_{t-2}$$

where $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with variance σ_z^2 .

(a) Is the process $\{X_t\}$ invertible? Justify your answer.

SOLUTION:

This is an $MA(2)$ process and can be written

$$X_t = \theta(B)Z_t$$

where

$$\theta(B) = 1 + \frac{1}{4}B - \frac{1}{8}B^2.$$

The process is invertible if the roots of

$$\theta(z) = 0$$

lie outside the unit circle. Therefore we solve

$$\begin{aligned} 1 + \frac{1}{4}z - \frac{1}{8}z^2 &= 0 \\ \implies \left(1 + \frac{1}{2}z\right) \left(1 - \frac{1}{4}z\right) &= 0 \end{aligned}$$

and the roots are $z = -2$ and $z = 4$, each of which lies outside the unit circle. Hence the process is invertible.

(b) Derive the autocovariance generating function (AGF) of this process.

SOLUTION:

Recall that the AGF is given by

$$\Gamma(z) = \sigma_z^2 \theta(z) \theta(z^{-1})$$

since

$$X_t = \theta(B)Z_t$$

is the process written as a general linear process. Then

$$\begin{aligned}\Gamma(z) &= \sigma^2 \left(1 + \frac{1}{4}z - \frac{1}{8}z^2\right) \left(1 + \frac{1}{4z} - \frac{1}{8z^2}\right) \\ &= \sigma_z^2 \left[1 + \frac{1}{16} + \frac{1}{64} + \left(\frac{1}{4} - \frac{1}{32}\right)z - \frac{1}{8}z^2 + \left(\frac{1}{4} - \frac{1}{32}\right)\frac{1}{z} - \frac{1}{8z^2}\right] \\ &= \sigma_z^2 \left[-\frac{1}{8z^2} + \frac{7}{32z} + \frac{69}{64} + \frac{7}{32}z - \frac{1}{8}z^2\right].\end{aligned}$$

(c) Using your answer to (b), determine the autocorrelation function of the process $\{X_t\}$.

SOLUTION:

We note that the autocovariance at lag h is given by the coefficient of z^h in the AGF. Hence, the autocovariance function is

$$\gamma(h) = \begin{cases} \frac{69\sigma_z^2}{64} & h = 0; \\ \frac{7\sigma_z^2}{32} & h = \pm 1; \\ -\frac{\sigma_z^2}{8} & h = \pm 2; \\ 0 & \text{otherwise.} \end{cases}$$

Since $\rho(h) = \gamma(h)/\gamma(0)$, it follows that the ACF is

$$\rho(h) = \begin{cases} 1 & h = 0; \\ \frac{14}{69} & h = \pm 1; \\ -\frac{8}{69} & h = \pm 2; \\ 0 & \text{otherwise.} \end{cases}$$

2. The process $\{Y_t : t \in \mathbb{Z}\}$ is defined

$$Y_t = \beta_0 + \beta_1 t + X_t$$

where $X_t = Z_t + 0.6Z_{t-1}$ and $\{Z_t\}$ is a white noise process with variance σ_z^2 .

(a) Determine $\mathbb{E}(Y_t)$.

SOLUTION:

$$\begin{aligned}\mathbb{E}(Y_t) &= \mathbb{E}(\beta_0 + \beta_1 t + X_t) \\ &= \beta_0 + \beta_1 t + \mathbb{E}(X_t) \\ &= \beta_0 + \beta_1 t + \mathbb{E}(Z_t + 0.6Z_{t-1}) \\ &= \beta_0 + \beta_1 t\end{aligned}$$

- (b) Determine the autocovariance function, $\gamma_Y(h)$, for the process $\{Y_t\}$.

SOLUTION:

Consider $\gamma_Y(h) = \text{Cov}(Y_t, Y_{t+h})$ for some integer h .

$$\begin{aligned}
 \gamma_Y(h) &= \mathbb{E}(Y_t Y_{t+h}) - \mathbb{E}(Y_t) \mathbb{E}(Y_{t+h}) \\
 &= \mathbb{E}[(\beta_0 + \beta_1 t + X_t)(\beta_0 + \beta_1(t+h) + X_{t+h})] - (\beta_0 + \beta_1 t)(\beta_0 + \beta_1(t+h)) \\
 &= (\beta_0 + \beta_1 t)(\beta_0 + \beta_1(t+h)) + (\beta_0 + \beta_1 t) \mathbb{E}(X_{t+h}) + (\beta_0 + \beta_1(t+h)) \mathbb{E}(X_t) \\
 &\quad + \mathbb{E}(X_t X_{t+h}) - (\beta_0 + \beta_1 t)(\beta_0 + \beta_1(t+h)) \\
 &= \mathbb{E}(X_t X_{t+h}) \\
 &= \mathbb{E}(Z_t + 0.6Z_{t-1})(Z_{t+h} + 0.6Z_{t+h-1}) \\
 &= \mathbb{E}(Z_t Z_{t+h}) + 0.6[\mathbb{E}(Z_{t-1} Z_{t+h}) + \mathbb{E}(Z_t Z_{t+h-1})] + 0.36\mathbb{E}(Z_{t-1} Z_{t+h-1})
 \end{aligned}$$

When $h = 0$

$$\begin{aligned}
 \text{Cov}(Y_t, Y_{t+h}) &= \mathbb{E}(Z_t^2) + 0.36\mathbb{E}(Z_{t-1}^2) \\
 &= 1.36\sigma_z^2.
 \end{aligned}$$

When $h = \pm 1$

$$\begin{aligned}
 \text{Cov}(Y_t, Y_{t+h}) &= 0.6\mathbb{E}(Z_t^2) \\
 &= 0.6\sigma_z^2.
 \end{aligned}$$

When $|h| > 2$

$$\text{Cov}(Y_t, Y_{t+h}) = 0.$$

Hence

$$\gamma_Y(h) = \begin{cases} 1.36\sigma_z^2 & h = 0; \\ 0.6\sigma_z^2 & h = \pm 1; \\ 0 & \text{otherwise.} \end{cases}$$

- (c) The process $\{W_t\}$ is defined

$$W_t = Y_t - Y_{t-1} - \beta_1.$$

What type of process is $\{W_t\}$? Is this process stationary and/or invertible?

SOLUTION:

We see that

$$\begin{aligned}
 W_t &= Y_t - Y_{t-1} - \beta_1 \\
 &= \beta_0 + \beta_1 t + X_t - (\beta_0 + \beta_1(t-1) + X_{t-1}) - \beta_1 \\
 &= X_t - X_{t-1} \\
 &= (Z_t + 0.6Z_{t-1}) - (Z_{t-1} + 0.6Z_{t-2}) \\
 &= Z_t - 0.4Z_{t-1} - 0.6Z_{t-2}.
 \end{aligned}$$

Clearly $\{W_t\}$ is an $MA(2)$ process. By definition, the process is stationary. To determine whether or not the process is invertible, we write the process as

$$W_t = (1 - 0.4B - 0.6B^2)Z_t$$

and check if the roots of

$$1 - 0.4z - 0.6z^2 = 0$$

lie outside the unit circle.

$$\begin{aligned} 1 - 0.4z - 0.6z^2 &= 0 \\ \implies z &= \frac{0.4 \pm \sqrt{0.4^2 + 2.4}}{-1.2} \\ &= 1 \text{ or } -\frac{5}{3}. \end{aligned}$$

One of the roots does not lie outside the unit circle and, hence, the process is not invertible.

3. A sequential sample of 100 observations is made from a time series. The sample autocorrelations r_j ($j = 1, \dots, 10$), calculated using this sample of data, are shown in the table below

j	1	2	3	4	5	6	7	8	9	10
r_j	0.31	0.37	-0.05	0.06	0.22	0.11	0.08	0.05	0.12	0.01

Can you suggest a plausible $ARMA$ process from which this sample of data may have arisen? Explain your answer.

SOLUTION:

Recall that if we assume that the sample autocorrelations are, approximately, normally distributed, then – if the process is white noise – we should expect all sample autocorrelations to lie within the interval $\left(-\frac{2}{\sqrt{n}}, \frac{2}{\sqrt{n}}\right)$. Here $n = 100$ and our interval of interest is

$$\left(-\frac{2}{10}, \frac{2}{10}\right) = (-0.2, 0.2).$$

Looking at the sample autocorrelations, we see that $|r_1|$ and $|r_2|$ and $|r_5|$ are all greater than 0.2 and all other values $|r_j|$ lie in the interval $(-0.2, 0.2)$. That r_1 and r_2 are fairly similar in magnitude indicates that the sample ACF is not declining exponentially to zero. As a result, we might guess that we have a pure MA process here.

Because $|r_5| > 0.2$ we might suspect that the process is an $MA(5)$ process. However, since both r_3 and r_4 lie close to zero, this is unlikely (and also note that $r_5 = 0.22$ lies close to 0.2). Overall, based on these sample autocorrelations, we'd argue that the sample of data has arisen from an $MA(2)$ process.

4. Suppose that the time series generated by the $MA(1)$ process

$$X_t = (1 - \theta B)Z_t \quad (1)$$

is incorrectly identified as the $AR(1)$ process

$$(1 - \phi B)X_t = W_t \quad (2)$$

where $\{Z_t\}$ and $\{W_t\}$ are independent white noise processes, each with variance σ^2 . Assume that the model (2) is erroneously fitted to data generated by model (1). Discuss the form of the model residuals.

SOLUTION:

The ‘true’ model is

$$X_t = (1 - \theta B)Z_t$$

which is an $MA(1)$ model. The incorrect fitted $AR(1)$ model may be written

$$(1 - \phi B)X_t = Z'_t$$

where $\{Z'_t\}$ is the residual process from the incorrect model. We see that

$$\begin{aligned} Z'_t &= (1 - \phi B)X_t \\ &= (1 - \phi B)(1 - \theta B)Z_t \\ &= (1 - \phi B - \theta B + \phi\theta B^2)Z_t \\ &= Z_t - (\phi + \theta)Z_{t-1} + \phi\theta Z_{t-2}. \end{aligned}$$

So $\{Z'_t\}$ is an $MA(2)$ process where $\theta_1 = -(\phi + \theta)$ and $\theta_2 = \phi\theta$.

5. The model

$$X_t = (1 - 0.6B)Z_t$$

is fitted to a sample of time series data, where $\{Z_t\}$ is a white noise process with variance σ_z^2 . The residuals from the model appear to look like an $AR(1)$ model of the form

$$(1 - 0.4B)Y_t = W_t$$

where $\{W_t\}$ is a white noise process. What type of model would you suggest should be fitted next? Justify your answer.

SOLUTION:

Since the residual process $\{Z'_t\}$ appears to be an $AR(1)$ process with

$$(1 - 0.4B)Z'_t = Z_t$$

where $\{Z_t\}$ is a white noise process. Then

$$Z'_t = (1 - 0.4B)^{-1}Z_t$$

and since $X_t = (1 - 0.6B)Z_t$ it follows that

$$\begin{aligned} X_t &= (1 - 0.6B)(1 - 0.4B)^{-1}Z_t \\ \implies (1 - 0.4B)X_t &= (1 - 0.6B)Z_t \end{aligned}$$

which is an $ARMA(1, 1)$ model with $\phi_1 = 0.4$ and $\theta_1 = -0.6$. Hence we should fit an $ARMA(1, 1)$ model to the time series $\{X_t\}$.

6. Suppose that $\{X_t\}$ is a weakly stationary process with autocovariance function $\gamma_X(h)$. The sample mean of n successive observations of the process is defined

$$\bar{X}_n = \frac{1}{n} \sum_{t=1}^n X_t.$$

Determine the variance of $\bar{X}_n$, expressing your answer in as simple a form as possible.

SOLUTION:

$$\begin{aligned} \text{Var}(\bar{X}_n) &= \text{Var}\left(\frac{1}{n} \sum_{t=1}^n X_t\right) \\ &= \frac{1}{n^2} \left[\sum_{t=1}^n \text{Var}(X_t) + \sum_{s \neq t} \text{Cov}(X_t, X_s) \right] \\ &= \frac{1}{n^2} \left[n\gamma_X(0) + 2 \sum_{r=1}^{n-1} \sum_{t=1}^{n-r} \text{Cov}(X_t, X_{t+r}) \right] \\ &= \frac{1}{n^2} \left[n\gamma_X(0) + 2 \sum_{r=1}^{n-1} \sum_{t=1}^{n-r} \gamma_X(r) \right] \\ &= \frac{1}{n^2} \left[n\gamma_X(0) + 2 \sum_{r=1}^{n-1} (n-r)\gamma_X(r) \right] \\ &= \frac{1}{n^2} \left[\sum_{r=-(n-1)}^{n-1} (n-|r|)\gamma_X(r) \right] \\ &= \frac{1}{n} \sum_{r=-(n-1)}^{n-1} \left(1 - \frac{|r|}{n}\right) \gamma_X(r). \end{aligned}$$

7. Consider the $AR(2)$ process

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$$

where $\{Z_t\}$ is a white noise process with variance σ_z^2 . Suppose that $\hat{\boldsymbol{\phi}} = (\hat{\phi}_1, \hat{\phi}_2)^\top$ denote estimators of the parameters $\boldsymbol{\phi} = (\phi_1, \phi_2)^\top$ formed by solving the Yule-Walker equations where a sample of size n of $\{X_t\}$ is taken.

Assuming that the $AR(2)$ process is (weakly) stationary, prove that

$$\text{Var} \left(n^{\frac{1}{2}} (\hat{\phi}_j - \phi_j) \right) \approx 1 - \phi_2^2 \text{ for } j \in \{1, 2\}$$

when n is large.

NOTE: You may assume that

$$n^{\frac{1}{2}} (\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}) \xrightarrow{D} \mathcal{N}_2(\mathbf{0}, (1 - \boldsymbol{\rho}_2^\top \boldsymbol{\phi}) \mathbf{P}_2^{-1})$$

where

$$\boldsymbol{\rho}_2 = (\rho_1, \rho_2)^\top; \quad \mathbf{P}_2 = \begin{bmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{bmatrix},$$

as $n \rightarrow \infty$.

SOLUTION:

From lectures, where $p = 2$, the Yule-Walker equations are given by

$$\begin{pmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} = \begin{pmatrix} \rho_1 \\ \rho_2 \end{pmatrix}.$$

Since

$$\begin{pmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{pmatrix}^{-1} = \frac{1}{1 - \rho_1^2} \begin{pmatrix} 1 & -\rho_1 \\ -\rho_1 & 1 \end{pmatrix},$$

it follows that

$$\begin{aligned} \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} &= \frac{1}{1 - \rho_1^2} \begin{pmatrix} 1 & -\rho_1 \\ -\rho_1 & 1 \end{pmatrix} \begin{pmatrix} \rho_1 \\ \rho_2 \end{pmatrix} \\ &= \frac{1}{1 - \rho_1^2} \begin{pmatrix} \rho_1(1 - \rho_2) \\ \rho_2 - \rho_1^2 \end{pmatrix}. \end{aligned}$$

Using the hint,

$$\begin{aligned} (1 - \boldsymbol{\rho}_2^\top \boldsymbol{\phi}) \mathbf{P}_2^{-1} &= \left(1 - (\rho_1 \ \rho_2) \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} \right) \frac{1}{1 - \rho_1^2} \begin{pmatrix} 1 & -\rho_1 \\ -\rho_1 & 1 \end{pmatrix} \\ &= \frac{1 - \rho_1 \phi_1 - \rho_2 \phi_2}{1 - \rho_1^2} \begin{pmatrix} 1 & -\rho_1 \\ -\rho_1 & 1 \end{pmatrix}. \end{aligned}$$

Each diagonal element of the above matrix is $\frac{1-\rho_1\phi_1-\rho_2\phi_2}{1-\rho_1^2}$ and this implies that, asymptotically,

$$\begin{aligned}
\text{Var} \left(\sqrt{n} \left(\hat{\phi}_1 - \phi_1 \right) \right) &= \text{Var} \left(\sqrt{n} \left(\hat{\phi}_2 - \phi_2 \right) \right) = \frac{1 - \rho_1\phi_1 - \rho_2\phi_2}{1 - \rho_1^2} \\
&= \frac{1 - \rho_1^2}{1 - \rho_1^2} + \frac{-(1 - \rho_1^2) + 1 - \rho_1\phi_1 - \rho_2\phi_2}{1 - \rho_1^2} \\
&= 1 + \frac{1}{1 - \rho_1^2} \left[\rho_1^2 - \rho_1 \frac{\rho_1(1 - \rho_2)}{1 - \rho_1^2} - \rho_2 \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2} \right] \\
&= 1 + \frac{1}{1 - \rho_1^2} \left[\frac{\rho_1^2 - \rho_1^4 - \rho_1^2 + \rho_1^2\rho_2 - \rho_2^2 + \rho_1^2\rho_2}{1 - \rho_1^2} \right] \\
&= 1 - \frac{(\rho_1^4 - 2\rho_1^2\rho_2 + \rho_2^2)}{(1 - \rho_1^2)^2} \\
&= 1 - \left[\frac{\rho_2 - \rho_1^2}{1 - \rho_1^2} \right]^2 \\
&= 1 - \phi_2^2.
\end{aligned}$$

8. Suppose $\{X_t\}$ is a (weakly) stationary process with mean zero and we define

$$D_m = \mathbb{E} \left[(X_t - \mathbf{X}_m^\top \boldsymbol{\beta}_m)^2 \right]$$

where

$$\begin{aligned}
\mathbf{X}_m &= (X_{t-1}, \dots, X_{t-m})^\top; \\
\boldsymbol{\beta}_m &= (\beta_{m,1}, \dots, \beta_{m,m})^\top.
\end{aligned}$$

- (a) By setting the derivatives of D_m with respect to $\beta_{m,1}, \dots, \beta_{m,m}$ to zero, show that finding $\boldsymbol{\beta}_m$ to minimise D_m is equivalent to finding $\boldsymbol{\phi} = (\phi_{m,1}, \dots, \phi_{m,m})^\top$ to solve the $m \times m$ system of Yule-Walker equations

$$\mathbf{P}_m \boldsymbol{\phi}_m = \boldsymbol{\rho}_m;$$

in the sense that $\boldsymbol{\beta}_m = \boldsymbol{\phi}_m$.

Here $\mathbf{P}_m$ is the $m \times m$ matrix whose $(i, j)^{\text{th}}$ element is ρ_{i-j} and $\boldsymbol{\rho} = (\rho_1, \dots, \rho_m)^\top$ with $\rho_h = \text{Cor}(X_t, X_{t+h})$.

SOLUTION:

Without loss of generality, assume that the mean of the process $\{X_t\}$ is zero (a similar argument to that given below can be followed if $\mathbb{E}(X_t) = \mu$ by considering

$Y_t = X_t - \mu$. Then, for $j = 1, \dots, m$

$$\begin{aligned}
\frac{\partial D_m}{\partial \beta_{m,j}} &= \frac{\partial}{\partial \beta_{m,j}} \mathbb{E} \left[(X_t - \mathbf{X}_m^\top \boldsymbol{\beta}_m)^2 \right] \\
&= \mathbb{E} \left[\frac{\partial}{\partial \beta_{m,j}} (X_t - \mathbf{X}_m^\top \boldsymbol{\beta}_m)^2 \right] \quad (\text{under regularity conditions}) \\
&= \mathbb{E} \left[-2X_{t-j} (X_t - \mathbf{X}_m^\top \boldsymbol{\beta}_m) \right] \\
&= -2 \left[\mathbb{E}(X_{t-j}X_t) - \sum_{k=1}^m \mathbb{E}(X_{t-j}X_{t-k})\beta_{m,k} \right] \\
&= -2 \left[\gamma(j) - \sum_{k=1}^m \gamma(j-k)\beta_{m,k} \right].
\end{aligned}$$

Setting $\frac{\partial D_m}{\partial \beta_{m,j}} = 0$, we obtain

$$\gamma(j) = \sum_{k=1}^m \gamma(j-k)\beta_{m,k}$$

and dividing the above by $\gamma(0)$ yields

$$\rho(j) = \sum_{k=1}^m \rho(j-k)\beta_{m,k}.$$

Noting that $\rho(j) = \rho(-j)$ we see that the above equation is similar to the j^{th} equation in the set of linear equations

$$\mathbf{P}_m \boldsymbol{\beta}_m = \boldsymbol{\rho}_m.$$

We know this set of equations as the Yule-Walker equations and it follows that the $\boldsymbol{\beta}_m$ that minimises D_m is also the solution to the Yule-Walker equations. Note: we omitted taking the second derivatives of D_m to check that $\boldsymbol{\beta}_m$ minimises D_m but the hessian matrix is positive definite and $\boldsymbol{\beta}_m$ does minimise D_m .

- (b) Hence deduce that $\beta_{m,m}$, the partial autocorrelation function at lag m , is equal to zero for an $AR(p)$ model if $m > p$.

SOLUTION:

Consider any stationary $AR(p)$ model

$$X_t = \phi_{p,1}X_{t-1} + \dots + \phi_{p,p}X_{t-p} + Z_t$$

where

$$\boldsymbol{\phi}_p = (\phi_{p,1}, \dots, \phi_{p,p})^\top$$

is the vector of $AR(p)$ coefficients.

If $m > p$ then the above model can be written as an $AR(m)$ model such that

$$X_t = \phi_{m,1}X_{t-1} + \dots + \phi_{m,m}X_{t-m} + Z_t$$

where

$$\boldsymbol{\phi}_m = (\phi_{m,1}, \dots, \phi_{m,m})^\top$$

is the vector of coefficients of the $AR(m)$ model and

$$\phi_{m,j} = \begin{cases} \phi_{p,j} & j = 1, \dots, p; \\ 0 & j = p+1, \dots, m. \end{cases}$$

From lectures, we know that $\boldsymbol{\phi}_m$ must satisfy the Yule-Walker equations

$$\mathbf{P}_m \boldsymbol{\phi}_m = \boldsymbol{\rho}_m$$

and this implies that

$$\boldsymbol{\phi}_m = \mathbf{P}_m^{-1} \boldsymbol{\rho}_m.$$

Furthermore, where $m > p$ we know that the m^{th} element of $\boldsymbol{\phi}_m$ is zero. That is

$$\phi_{m,m} = [\mathbf{P}_m^{-1} \boldsymbol{\rho}_m]_m = 0.$$

From lectures, we know that the PACF at lag m is given by

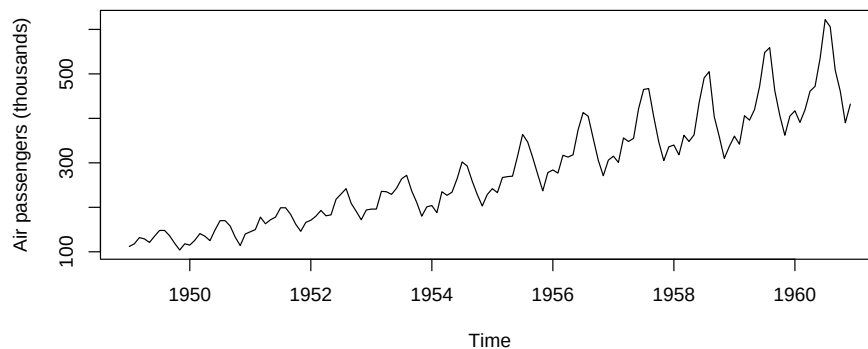
$$\beta_{m,m} = [\mathbf{P}_m^{-1} \boldsymbol{\rho}_m]_m = 0 \text{ if } m > p.$$

Hence the PACF of an $AR(p)$ process at lag m is equal to zero if $m > p$.

9. The dataset `USAirPassengers.csv` can be found on the course moodle page and contains monthly US Air Passenger numbers from 1949 to 1960, covered during Lecture 10. Using R, fit an appropriate time series model to these data and check the fit of your model.

SOLUTION:

We begin by producing a time plot of the data. There is obvious seasonality in the



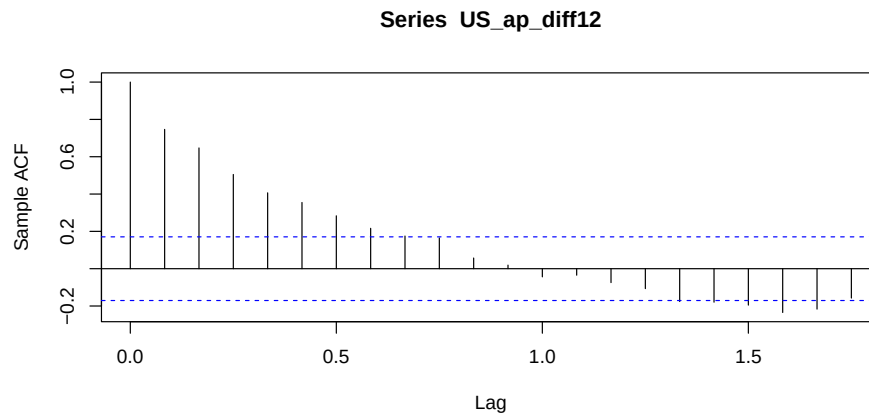
data. As the data have been collected monthly, we difference seasonally at lag 12. A time



plot of the seasonally differenced data is shown below.

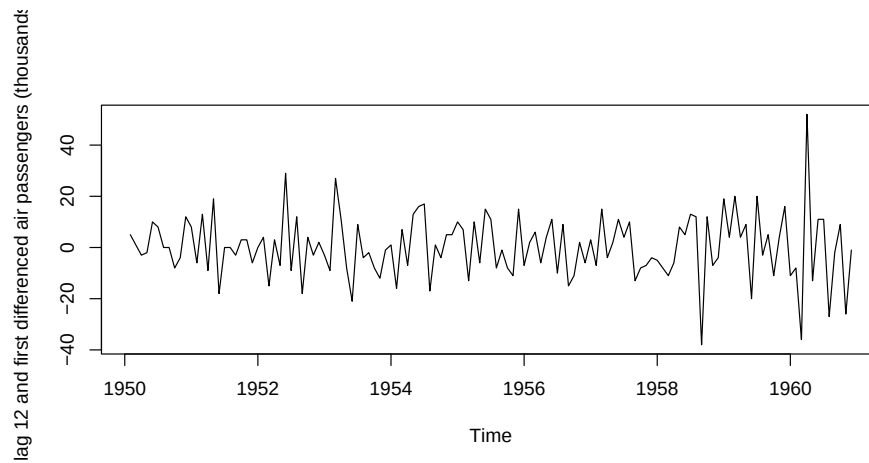
Although the seasonality appears to have been removed, the data do not seem to be stationary. We'll check this by producing a plot of the sample ACFs for the differenced (lag 12) series.

The plot of the sample ACF against the lag suggests that the data are still not sta-

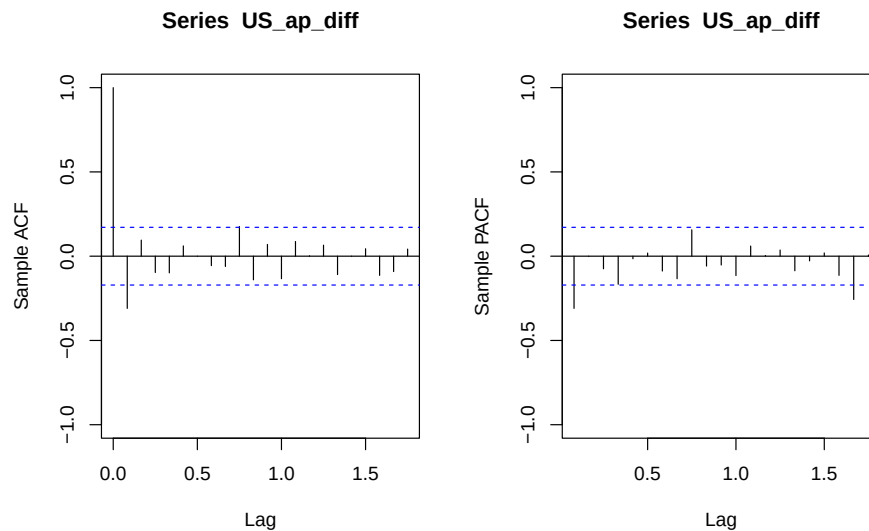


tionary. As a next step, we take the first difference of the seasonally differenced data. A time plot of these data is shown below.

These data certainly appear to be more stationary (the mean appears to be constant and equal to zero).



We produce a plot of the sample ACF and sample PACF against the lag for these data to examine this further (shown below).



At this point we'd assume that these differenced data are stationary. Note that if X_t is the original process then we will model the process

$$Y_t = (1 - B^{12})(1 - B)X_t$$

or, alternatively, we'll fit an $ARIMA(p, 1, q) \times (P, 1, Q)_{12}$ model to the original data.

Looking at the correlogram (sample ACF plot) we begin by fitting an $ARIMA(0, 1, 1) \times (0, 1, 0)_{12}$ model. The fitted model output is

Call:

```

arima(x = US_ap, order = c(0, 1, 1),
seasonal = list(order = c(0, 1, 0), period = 12))

```

Coefficients:

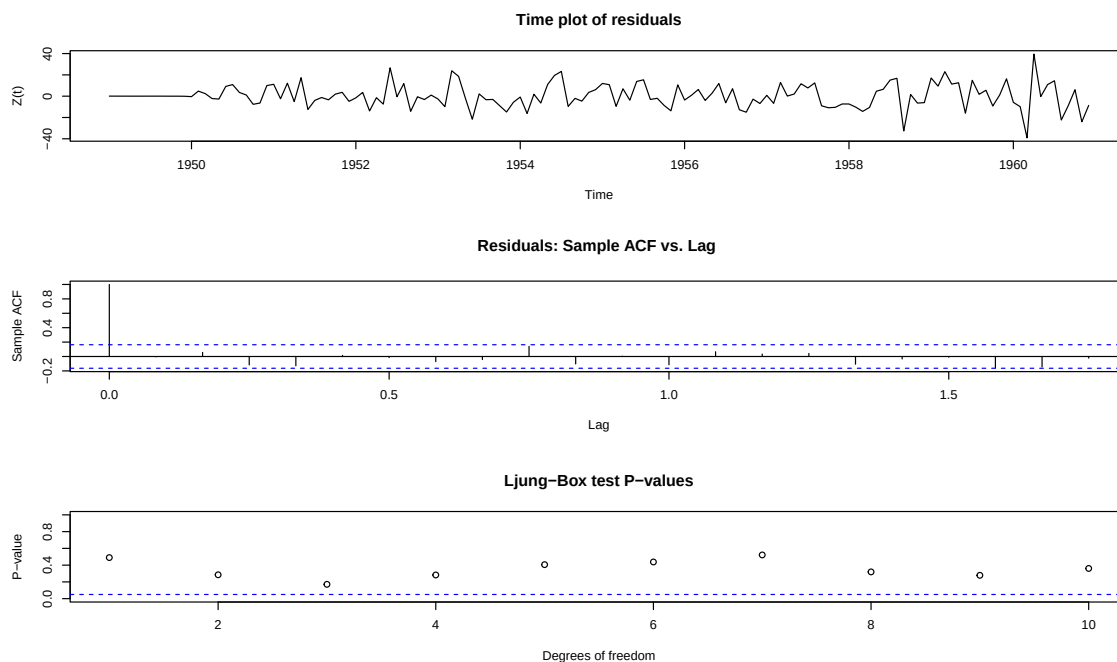
```

          ma1
        -0.3184
s.e.      0.0877

```

σ^2 estimated as 137.3: log likelihood = -508.32, aic = 1020.64

For the residuals, we produce a time plot, plot of the sample ACF against the lag and a plot of the Ljung-Box test statistics, shown below. Clearly, the $ARIMA(0, 1, 1) \times (0, 1, 0)_{12}$



could be a reasonably good fit. The residuals look like white noise (no correlation between residuals) and the Ljung-Box test statistics are all fairly large.

Let's explore whether an AR component should be included. We'll try fitting an $ARIMA(1, 1, 1) \times (0, 1, 0)_{12}$ next. The fit of the model is

Call:

```

arima(x = US_ap, order = c(1, 1, 1),
seasonal = list(order = c(0, 1, 0), period = 12))

```

Coefficients:

```

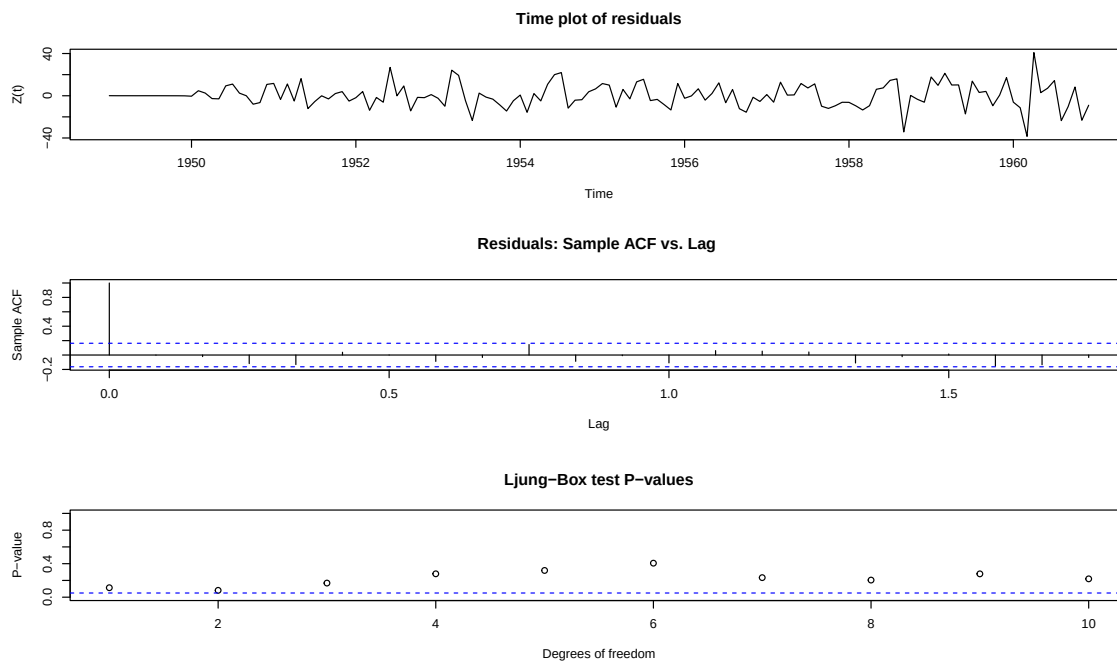
          ar1          ma1
        -0.3009        -0.0073

```

s.e. 0.3835 0.4133

sigma^2 estimated as 137: log likelihood = -508.2, aic = 1022.39

For the residuals, we produce a time plot, plot of the sample ACF against the lag and a plot of the Ljung-Box test statistics, shown below.



This model looks like a good fit to the data, however, the AIC is higher than that for the MA(1) model and this model contains an extra parameter. At this point, we might fit the MA(1) model to the data.

It is worth noting that the $ARIMA(2, 1, 1) \times (0, 1, 0)_{12}$ might be a better model. The R output for this model is shown below. Note the smaller AIC value than that for the MA(1) model. However, this model contains two extra parameters when compared to the MA(1) model so it may not be preferable to the MA(1) model.

Call:

```
arima(x = US_ap, order = c(2, 1, 1),  
seasonal = list(order = c(0, 1, 0), period = 12))
```

Coefficients:

	ar1	ar2	ma1
	0.5960	0.2143	-0.9819
s.e.	0.0888	0.0880	0.0292

sigma² estimated as 129.3: log likelihood = -504.92, aic = 1017.85